

# CSE 485 Midterm Exam

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## I. QUESTION

- Briefly describe the difference between classification, prediction and clustering.
- Give three possible strategies for filling in missing values.

## II. QUESTION

Draw appropriate schemas for the following data warehouses.

- The data warehouse consists of three dimensions time, doctor and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.
- The data warehouse for Big-University consists of the following four dimensions: student, course, semester, and instructor, and two measures count and avg\_grade.

## III. QUESTION

Given the following error rates for a four feature problem, what will be the result of the following attribute subset selection procedures?

- Stepwise forward selection
- Stepwise backward elimination

Error Rates:  $(F_1)$ , 0.23;  $(F_2)$ , 0.25;  $(F_3)$ , 0.36;  $(F_4)$ , 0.24;  $(F_1, F_2)$ , 0.18;  $(F_1, F_3)$ , 0.19;  $(F_1, F_4)$ , 0.20;  $(F_2, F_3)$ , 0.16;  $(F_2, F_4)$ , 0.29;  $(F_3, F_4)$ , 0.22;  $(F_1, F_2, F_3)$ , 0.39;  $(F_1, F_2, F_4)$ , 0.45;  $(F_1, F_3, F_4)$ , 0.24;  $(F_2, F_3, F_4)$ , 0.36;  $(F_1, F_2, F_3, F_4)$ , 0.59.

## IV. QUESTION

Suppose that the data for analysis include the attribute age. The age values for data tuples are: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. According to this data,

- Use min-max normalization to transform the value 35 for age onto the range [0.0, 1.0].
- What are the mean, the median and the mode?
- Plot an equiwidth histogram of width 10.

## V. QUESTION

The following table consists of training data from an employee database. Let status be the class label attribute.

| Department | Status | Age    | Salary   |
|------------|--------|--------|----------|
| sales      | senior | 31..35 | 46K..50K |
| sales      | junior | 26..30 | 26K..30K |
| sales      | junior | 31..35 | 31K..35K |
| systems    | junior | 21..25 | 46K..50K |
| systems    | senior | 31..35 | 66K..70K |
| systems    | junior | 26..30 | 46K..50K |
| systems    | senior | 41..45 | 66K..70K |
| marketing  | senior | 36..40 | 46K..50K |
| marketing  | junior | 31..35 | 31K..35K |
| secretary  | senior | 41..45 | 31K..35K |
| secretary  | junior | 26..30 | 26K..30K |

Which attribute is most relevant?

## VI. QUESTION

A database has four transactions. Let min\_sup = 60% and min\_conf = 80%.

| TID  | date     | items_bought    |
|------|----------|-----------------|
| T100 | 10/15/99 | {K, A, D, B}    |
| T200 | 10/15/99 | {D, A, C, E, B} |
| T300 | 10/19/99 | {C, A, B, E}    |
| T400 | 10/22/99 | {B, A, D}       |

Find all frequent itemsets and association rules using Apriori algorithm.